

Geometry-MMALS G1 v1.0.4

Matched Initialization, Matched Compute, and Retention Anchors

Archive status	Protocol correction report and executable development design.
Accepted claim	C0 pipeline/protocol integrity only.
Not claimed	C1-C6 qualification, quantum advantage, backward transfer, replay-free continual learning, or domain-general geometry.

Executive decision

v1.0.4 isolates the geometry objective from extra compute and random initialization, then tests a no-extra-forward retention anchor.

The v1.0.3 cross-angle paired experiment was the first informative Geometry-MMALS G1 run. It showed stronger local route/context ordering and improved average interpolation, but the comparison remained confounded by different MMALS initializations and unequal forward-compute budgets. It also improved plasticity while increasing forgetting on earlier angles.

The primary v1.0.4 contrast is now **paired_geometry** versus **paired_compute_matched_no_geometry**. These methods share the same multi-angle tensors, immutable initial state, source order, image forwards, and optimizer steps. A separate **paired_geometry_anchor_010** method tests whether old-angle classification evidence can reduce forgetting without extra image-forward compute.

Research history

Version	Main question	Outcome
v1.0.0	Can functional geometry be specified and falsified?	Article, evidence ladder, C0 scaffold.
v1.0.1	Can claims, gates, notation, and memory contracts be audited?	C0-only status and pilot gates.
v1.0.2	Can route loss and Colab execution remain finite?	Numerical stability passed.
v1.0.3	Does optimization contain real cross-angle evidence?	Same-source paired protocol; first informative run.
v1.0.4	Does geometry add value at matched initialization/compute, and can an anchor reduce forgetting?	Implemented; qualification pending.

v1.0.3 evidence carried forward

Metric	No geometry	Paired geometry	Interpretation
Route source rho	0.378	0.492	Stronger local route order.
Context source rho	0.289	0.497	Stronger local context order.
Mean interpolation accuracy	0.278	0.346	Promising but asymmetric.
Mean forgetting	0.059	0.105	Stability cost increased.
Causal specificity ratio	~0.17	~0.66-0.74	Improved, still below one.

These values are development evidence only. Synthesis geometry did not improve, interpolation gains were directionally asymmetric, and causal specificity remained weaker than matched orthogonal directions.

Why the v1.0.4 correction is necessary

Initialization confound

Each v1.0.3 method created a fresh context network, router, host bank, and classifier. v1.0.4 builds one base model exactly once, records its SHA-256 digest, and reloads that state for every method.

Compute confound

The current-only reference processed one image per source, while paired geometry processed all seen angles. v1.0.4 adds a paired compute-matched control using the same multi-angle tensors and the same optimizer-step budget with geometry weight zero.

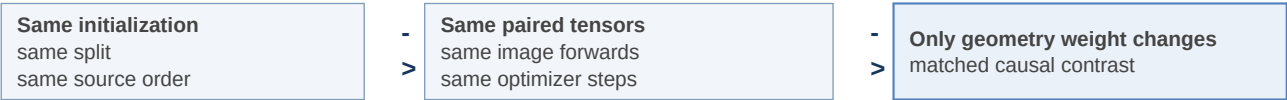
Stability-plasticity trade-off

Paired geometry improved final and held-out accuracy but increased forgetting. The anchor condition adds classification loss on old-angle views already present in the paired batch. This changes the loss but not the image-forward count.

Method matrix

Method	Stage inputs	Geo	Anchor	Role
current_only_no_geometry	Current angle	0	0	Cheaper external reference
paired_compute_matched_no_geometry	All seen angles	0	0	Primary matched control
paired_geometry	All seen angles	On	0	Geometry treatment
paired_geometry_anchor_010	All seen angles	On	0.10	Retention treatment

Attribution logic



Experimental invariants

- One deterministic source split is shared by all methods.
- One immutable MMALS initialization is shared and recorded by SHA-256.
- Paired variants use method-independent DataLoader seeds for every stage and epoch.
- Paired variants must match exactly on image forwards, source examples, and optimizer steps.
- Interpolation and extrapolation angles never enter training or model selection.
- The true angle is available only to the controlled loss and evaluator, never to the router.
- The sensory encoder is frozen before MMALS training.
- Qualification mode stops when the sensory acceptance gate fails.

Measurements and evidence bundle

- Stage losses, wall time, source counts, image-forward counts, and optimizer-step counts.
- Staged evaluation on trained, interpolation, and extrapolation angles.
- Trained-angle forgetting and final accuracy.
- Source-block and factor-centroid geometry for sensory, context, route, and synthesis spaces.
- Held-out route/context interpolation with nearest-route control.
- Route mass and host ablation mean, median, and positive-contribution fraction.
- Signed causal route effect, matched orthogonal effect, causal specificity, and class-evidence change.
- Method summaries plus explicit geometry-attribution and retention-anchor deltas.
- Immutable manifest with package version, git SHA, split hashes, initial-state hash, and non-claims.

Gate interpretation

Contrast	Treatment vs control	Expected evidence
Geometry effect at matched compute	paired_geometry vs paired_compute_matched_no_geometry	Positive route/context order delta, useful interpolation delta, acceptable synthesis/forgetting trade-off, budget audit passed.
Retention anchor effect	paired_geometry_anchor_010 vs paired_geometry	Lower forgetting without losing most geometry or interpolation benefit, and no extra image-forward compute.

Sensory acceptance boundary

The development notebook pretrains the sensory grove on a larger source subset than v1.0.3 and evaluates held-out 0-degree identity accuracy before MMALS training. Development mode warns below 0.85. Qualification mode stops below 0.95. This prevents a geometry claim from being built on an immature identity representation.

Remaining gate dependencies

- **C1:** frozen numeric thresholds and multi-seed confidence intervals.
- **C2:** matched-control interpolation gains across seeds and no held-out leakage.
- **C3:** cross-seed host-role matching and locality measures.
- **C4:** source-block bootstrap, identity preservation, and causal specificity above the frozen threshold.
- **C5:** explicit functional transport and dual-memory experiments.
- **C6:** demonstrated operational benefit at controlled cost.

Archive and execution plan

Recommended GitHub paths

```
notebooks/Geometry_MMALS_G1_MatchedCompute_Retention_v1_0_4.ipynb
docs/reports/Geometry_MMALS_G1_v1_0_4_Matched_Compute_Retention_Report.pdf
docs/reports/Geometry_MMALS_G1_v1_0_4_Matched_Compute_Retention_Report.md
docs/changes/Geometry_MMALS_G1_v1_0_3_to_v1_0_4.md
```

A completed run should additionally archive results/v1_0_4/seed_/ with the raw CSV tables and run_manifest.json.

Run sequence

- Run development mode first and inspect finite-value and sensory guards.
- Confirm the paired compute audit passes before interpreting metrics.
- Inspect the geometry attribution contrast before comparing against the cheaper current-only reference.
- Inspect the retention anchor contrast separately.
- Freeze thresholds and anchor weight before multi-seed qualification.
- Run at least five pilot seeds and ten final seeds.
- Add hardened continual-learning baselines only after the internal contrast is stable.

Tracked changes

- CHG-104-01 Shared MMALS initialization.
- CHG-104-02 Deterministic method-independent source order.
- CHG-104-03 Compute-matched paired control.
- CHG-104-04 Explicit geometry-attribution contrast.
- CHG-104-05 No-extra-forward retention anchor.
- CHG-104-06 Forward and optimizer-step accounting.
- CHG-104-07 Sensory acceptance gate.
- CHG-104-08 Initial-state and split digests.
- CHG-104-09 Budget assertions.
- CHG-104-10 Attribution and retention deltas.
- CHG-104-11 Clean archival evidence bundle.
- CHG-104-12 Strict C1-C6 non-claims.

Epistemic conclusion

v1.0.4 does not assert that Geometry-MMALS works. It creates the matched experimental contrast required to decide whether the v1.0.3 cross-angle signal came from the geometry objective itself, from extra compute, from different initialization, or from a stability-plasticity trade-off. The next acceptable scientific statement must be based on the matched paired contrast, not on the cheaper current-only baseline alone.